

Technical Summary:

Self-Consistency Improves Chain of Thought Reasoning in Language Models

Kai-Yu Lu

1 Research Problem and Motivation

1.1 Problem Setting

This work studies *reasoning at inference time* for large language models, focusing on tasks where a correct answer typically requires multiple intermediate steps rather than a single factual recall. A common inference technique for such tasks is *chain-of-thought prompting* (CoT), which provides a small set of exemplars that include step-by-step reasoning, encouraging the model to produce an explicit reasoning trace before the final answer.

1.2 Motivation and Research Gap

A standard decoding choice for CoT is *greedy decoding*, which deterministically selects the most likely next token at each generation step. Greedy decoding produces a single reasoning path, so any early mistake can propagate and determine the final answer. Alternative decoding methods such as *beam search* attempt to keep multiple partial continuations, but beam search is designed to optimize likelihood under the model and tends to reduce diversity of outputs, which is undesirable when multiple distinct correct reasoning routes exist.

The core gap addressed is that existing CoT inference pipelines typically select a single reasoning path, while difficult reasoning questions can admit multiple valid reasoning paths that converge to the same answer. The work hypothesizes that correct reasoning paths, although diverse in steps and phrasing, tend to agree on the final answer more often than incorrect paths.

2 Related Work

2.1 Chain-of-Thought Prompting

Chain-of-thought prompting (CoT) is an inference prompting strategy that elicits intermediate reasoning steps in the model output. CoT is effective for arithmetic, commonsense, and symbolic reasoning tasks, but its performance depends on both the prompt design and the decoding strategy used at inference time.

2.2 Decoding, Sampling, and Ensembling

Several inference-time techniques can increase performance without parameter updates:

- *Stochastic sampling* such as temperature sampling and top- k truncation can produce diverse outputs by sampling from a softened or restricted token distribution.

- *Beam search* keeps multiple high-likelihood continuations, but can reduce diversity and does not directly optimize answer agreement.
- *Prompt ensembling* aggregates predictions across multiple prompts or prompt permutations to mitigate prompt sensitivity.

This work positions self-consistency as a *single-model self-ensemble* that aggregates answers across multiple sampled CoT reasoning paths.

3 Dataset Construction

3.1 Scope

No new dataset is constructed. Evaluation is conducted on established reasoning benchmarks covering arithmetic, commonsense, and symbolic reasoning.

3.2 Evaluation Datasets and Characteristics

Table 1 summarizes the benchmark families and their task characteristics.

Category	Dataset	Task Definition	Metric
Arithmetic reasoning	AddSub; MultiArith; ASDiv; AQuA-RAT; SVAMP; GSM8K	Math word problems requiring multi-step numerical reasoning. Outputs are typically numeric.	Accuracy (%)
Commonsense reasoning	CommonsenseQA; StrategyQA; ARC-Easy; ARC-Challenge	Multiple-choice or yes/no questions requiring commonsense knowledge and multi-step justification.	Accuracy (%)
Symbolic reasoning	Last Letter Concatenation; Coin Flip	Rule-based symbolic transformations and state tracking, including out-of-distribution settings with longer compositions.	Accuracy (%)

Table 1: Evaluation datasets and task characteristics.

4 Query Protocol and Task Definitions

4.1 Prompting Protocol

All experiments use *few-shot prompting* without training or fine-tuning. The prompts match the referenced CoT prompting setup: arithmetic reasoning uses 8 manually written exemplars; commonsense reasoning uses 4 to 7 exemplars selected from training data with manually composed CoT prompts.

4.2 Key Definitions

Question denotes the input problem statement. **Prompt** denotes the concatenation of exemplars and the current question. A **reasoning path** is the generated chain-of-thought text that precedes the final answer. A **candidate answer** is the final extracted answer associated with a generated reasoning path.

4.3 Sampling Configuration and Reporting

Self-consistency is evaluated by sampling $m = 40$ candidate outputs per run, with results averaged over 10 runs. Sampling uses temperature sampling and, for some models, top- k truncation. Table 2 summarizes the reported sampling settings.

Component	Configuration
Runs and samples	10 runs; $m = 40$ independently sampled outputs per run
Sampling (UL2-20B, LaMDA-137B)	Temperature $T = 0.5$ with top- k truncation at $k = 40$
Sampling (PaLM-540B)	Temperature $T = 0.7$ with top- k truncation at $k = 40$
Sampling (GPT-3 engines)	Temperature $T = 0.7$ without top- k truncation
Baseline decoding	CoT prompting with greedy decoding

Table 2: Inference and sampling configuration.

5 Modeling Approach

5.1 Method Overview

Self-consistency replaces single-path greedy decoding with a three-stage inference procedure:

1. **Sample diverse reasoning paths:** sample m outputs from the model under a CoT prompt using stochastic decoding.
2. **Extract candidate answers:** parse the final answer from each sampled output.
3. **Aggregate by answer agreement:** select the answer that is most frequent across the m samples.

Intuitively, stochastic sampling explores multiple plausible solution routes, and majority voting leverages the tendency of correct routes to converge on the same answer.

5.2 Sequence Probability Definition

$$P(r_i, a_i \mid \text{prompt, question}) \triangleq \exp \left(\frac{1}{K} \sum_{k=1}^K \log P(t_k \mid \text{prompt, question}, t_{1:k-1}) \right) \quad (1)$$

Explanation of Eq. (1). r_i denotes the i -th sampled reasoning path, and a_i denotes its associated final answer. The generated output is tokenized into a length- K token sequence $(t_1, \dots, t_K)$, where $t_{1:k-1}$ is the prefix up to token $k - 1$. $P(t_k \mid \cdot)$ is the language model token probability of the next token given the prompt, question, and previously generated tokens. The inner sum is the total log-likelihood of the sequence; dividing by K performs length normalization so that longer outputs are not systematically penalized. The outer exponential maps normalized log-likelihood back to a positive score. In this work, this formulation supports comparing or weighting sampled outputs, though the primary self-consistency decision rule uses answer agreement.

5.3 Self-Consistency Answer Selection

$$a^* = \arg \max_a \sum_{i=1}^m \mathbb{I}(a_i = a) \quad (2)$$

Explanation of Eq. (2). m is the number of sampled outputs. a_i is the answer extracted from the i -th output. $\mathbb{I}(\cdot)$ is an indicator function that equals 1 if its argument is true and 0 otherwise. For any candidate answer a , the sum counts how many sampled outputs produced that answer. The $\arg \max$ selects the answer with the highest count, which implements majority vote. The role of Eq. (2) is to operationalize the hypothesis that correct reasoning paths have higher answer agreement than incorrect paths.

5.4 Aggregation Variants

The work also compares multiple aggregation strategies beyond majority vote, including likelihood-weighted averaging and weighted summation, with and without length normalization. Table 3 reports the accuracy impact of these strategies.

Aggregation Strategy	GSM8K	MultiArith	AQuA	SVAMP	CSQA	ARC-c
Greedy decode	56.5	94.7	35.8	79.0	79.0	85.2
Weighted avg (unnormalized)	56.3 ± 0.0	90.5 ± 0.0	35.8 ± 0.0	73.0 ± 0.0	74.8 ± 0.0	82.3 ± 0.0
Weighted avg (normalized)	22.1 ± 0.0	59.7 ± 0.0	15.7 ± 0.0	40.5 ± 0.0	52.1 ± 0.0	51.7 ± 0.0
Weighted sum (unnormalized)	59.9 ± 0.0	92.2 ± 0.0	38.2 ± 0.0	76.2 ± 0.0	76.2 ± 0.0	83.5 ± 0.0
Weighted sum (normalized)	74.1 ± 0.0	99.3 ± 0.0	48.0 ± 0.0	86.8 ± 0.0	80.7 ± 0.0	88.7 ± 0.0
Unweighted sum (majority vote)	74.4 ± 0.1	99.3 ± 0.0	48.3 ± 0.5	86.6 ± 0.1	80.7 ± 0.1	88.7 ± 0.1

Table 3: Accuracy comparison of different answer aggregation strategies on PaLM-540B. Metrics are accuracy (%).

6 Empirical Results

6.1 Arithmetic Reasoning Results

Table 4 reports arithmetic reasoning accuracy across multiple models and benchmarks. Self-consistency yields consistent improvements over CoT with greedy decoding, with especially large gains on harder datasets and smaller models.

Method	AddSub	MultiArith	ASDiv	AQuA	SVAMP	GSM8K
PaLM-540B (CoT prompting)	72.7	91.7	55.4	57.9	79.0	56.5
PaLM-540B (self-consistency)	91.2 (+18.5)	97.0 (+5.3)	79.6 (+24.2)	74.3 (+16.4)	86.8 (+7.8)	74.4 (+17.9)
LaMDA-137B (CoT prompting)	46.0	93.6	25.2	26.5	69.1	30.3
LaMDA-137B (self-consistency)	78.4 (+32.4)	98.0 (+4.4)	63.3 (+38.1)	53.1 (+26.6)	79.7 (+10.6)	49.7 (+19.4)
UL2-20B (CoT prompting)	22.7	85.9	7.8	4.2	44.9	3.0
UL2-20B (self-consistency)	57.0 (+34.3)	95.7 (+9.8)	17.2 (+9.4)	31.5 (+27.3)	53.9 (+9.0)	15.6 (+12.6)
GPT-3 (code-davinci-001, CoT prompting)	57.2	59.5	52.7	18.9	39.8	14.6
GPT-3 (code-davinci-001, self-consistency)	67.8 (+10.6)	82.7 (+23.2)	63.4 (+10.7)	23.2 (+4.3)	57.3 (+17.5)	28.1 (+13.5)
GPT-3 (code-davinci-002, CoT prompting)	68.2	84.0	56.2	23.2	54.4	38.9
GPT-3 (code-davinci-002, self-consistency)	75.5 (+7.3)	91.4 (+7.4)	62.1 (+5.9)	30.5 (+7.3)	59.4 (+5.0)	45.0 (+6.1)

Table 4: Arithmetic reasoning accuracy. Metrics are accuracy (%).

6.2 Comparison to Sample-and-Rank

The work compares self-consistency to sample-and-rank, a strategy that samples multiple outputs and selects the one with the highest model score. Reported trends indicate that, under the same number of sampled

outputs, self-consistency yields larger accuracy gains than sample-and-rank, emphasizing the importance of aggregating by answer agreement rather than selecting a single highest-scored sample.

6.3 Comparison to Beam Search

Table 5 compares self-consistency to beam search decoding on UL2-20B under matched numbers of beams or sampled paths. Self-consistency with sampling outperforms beam search on both AQuA and MultiArith at larger beam or path counts, supporting the claim that diversity in reasoning paths is central.

Method	1	5	10	20	40
AQuA accuracy (%)					
Beam search decoding (top beam)	23.6	19.3	16.1	15.0	10.2
Self-consistency using beam search	23.6	19.8 ± 0.3	21.2 ± 0.7	24.6 ± 0.4	24.2 ± 0.5
Self-consistency using sampling	19.7 ± 2.5	24.9 ± 2.6	25.3 ± 1.8	26.7 ± 1.0	26.9 ± 0.5
MultiArith accuracy (%)					
Beam search decoding (top beam)	10.7	12.0	11.3	11.0	10.5
Self-consistency using beam search	10.7	11.8 ± 0.0	11.4 ± 0.1	12.3 ± 0.1	10.8 ± 0.1
Self-consistency using sampling	9.5 ± 1.2	11.3 ± 1.2	12.3 ± 0.8	13.7 ± 0.9	14.7 ± 0.3

Table 5: Beam search versus self-consistency on UL2-20B. Beam size or number of sampled paths is shown as the column header. Metrics are accuracy (%).

6.4 Compatibility with Prompt Ensembling

Self-consistency can be combined with prompt ensembling strategies. Table 6 reports GSM8K results for PaLM-540B using 40 prompt variants or 40 sampled paths. The combination of self-consistency with multiple prompt sets yields the best reported value in this comparison.

Method	GSM8K accuracy (%)
Different sets of prompts ($\times 40$)	58.9
Prompt permutation ($\times 40$)	59.6
Self-consistency ($\times 40$)	74.4
Self-consistency + different sets of prompts ($\times 40$)	75.4
Self-consistency + prompt permutation ($\times 40$)	73.8

Table 6: Combining self-consistency with prompt ensembling strategies on GSM8K using PaLM-540B. Metrics are accuracy (%).

7 Summary

7.1 Key Innovations Relative to Prior Inference

- **Multi-path reasoning at inference time:** instead of selecting a single CoT output, the method samples many CoT reasoning paths and aggregates by answer agreement.
- **Answer-based self-ensembling:** Eq. (2) defines a simple, model-agnostic majority vote rule that improves performance without additional training.

- **Broad empirical validation:** improvements are reported across arithmetic, commonsense, and symbolic reasoning tasks and across multiple model families and scales.

7.2 Limitations and Future Directions

- **Increased inference cost:** generating m reasoning paths increases compute and latency relative to single-path greedy decoding.
- **Dependence on sampling diversity:** the approach relies on stochastic decoding to produce diverse reasoning paths; decoding configurations that reduce diversity can limit gains.
- **Failure under systematic bias:** if most sampled paths converge to the same incorrect answer due to a shared misconception, majority vote will select that incorrect answer.